Basic Science Department

BAS 141
Engineering Chemistry

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Chapter 1: Physical Chemistry

Lecture 1

List of Content

Part 1: Gases: Their Properties and Behavior

1.1. Gases and gas pressure

1.2. The gas laws

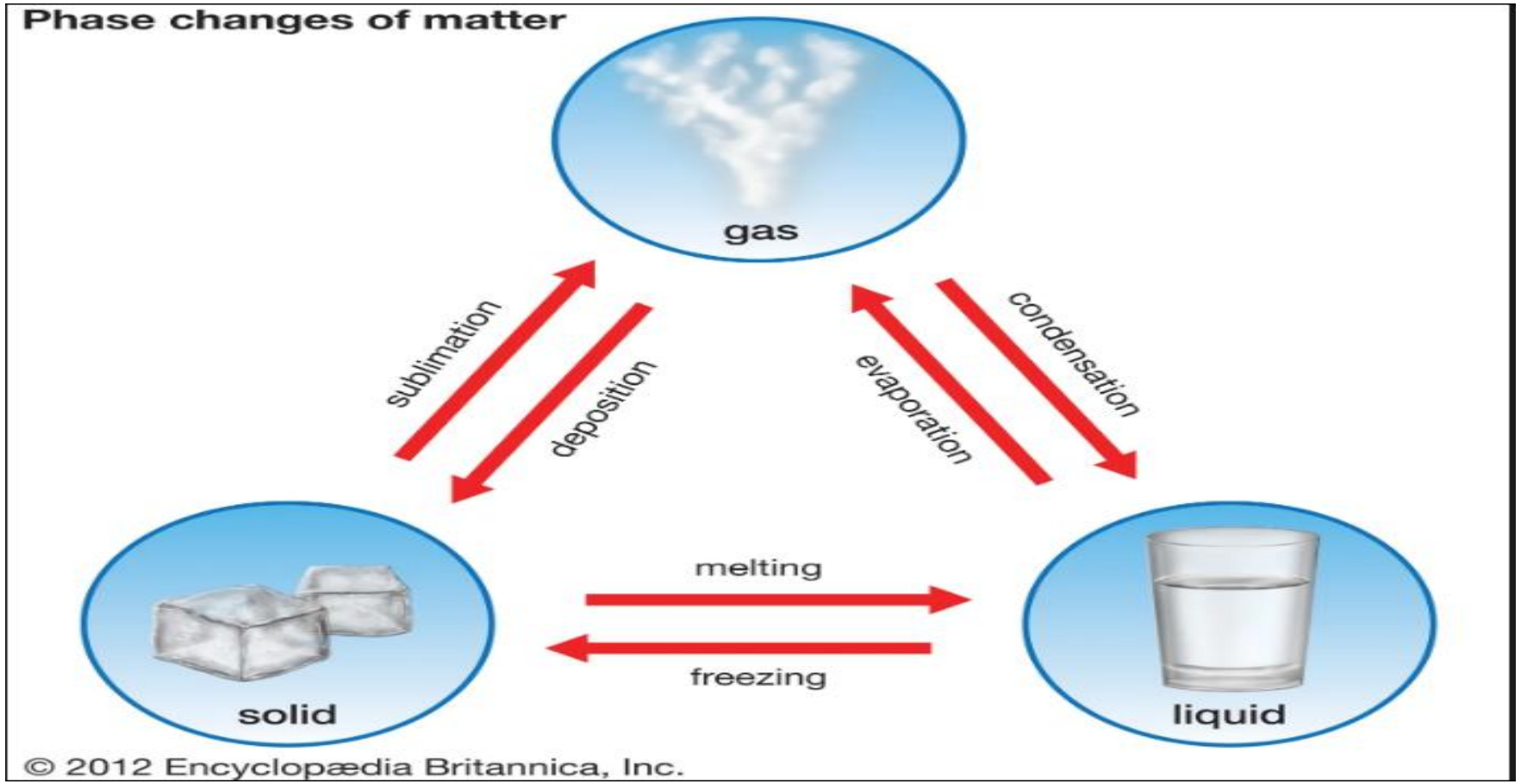
1.3. The ideal gas law

1.4.The kinetic molecular theory of gases

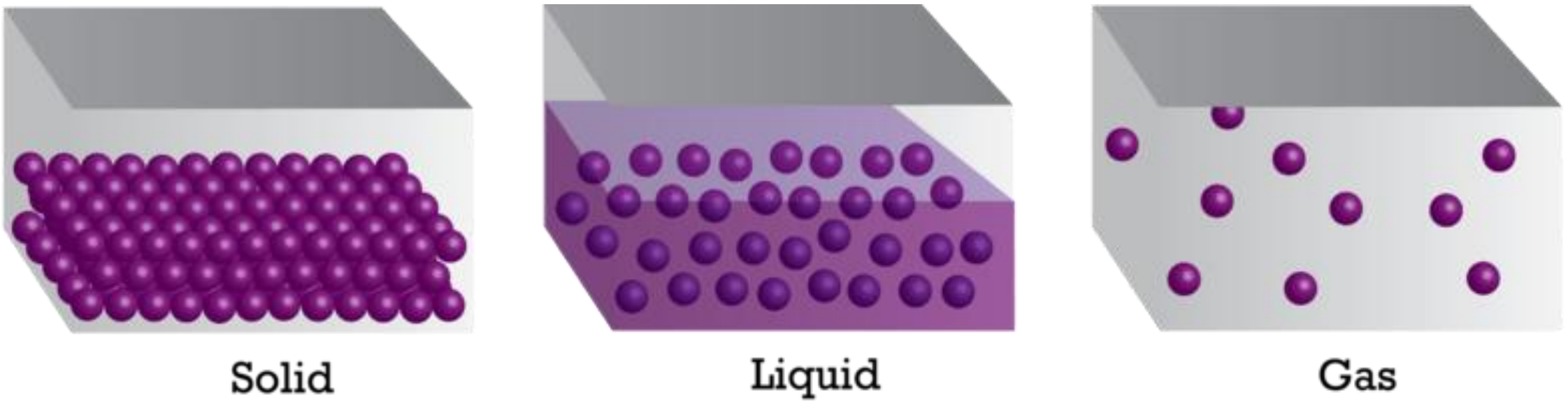
1.5. The behavior of real gases

Physical Chemistry

Branch of chemistry concerned with interactions and transformations of materials



Matter exists in three physical states: solid, liquid, and gas.


Solid
Liquid
Gas
Kinetic Energy
Particle Motions
Force of Attraction Between Particles



- Rigid
- Fixed Shape
- Fixed Volume
- Cannot be squashed



- Not Rigid
- No Fixed Shape
- Fixed Volume
- Cannot be squashed



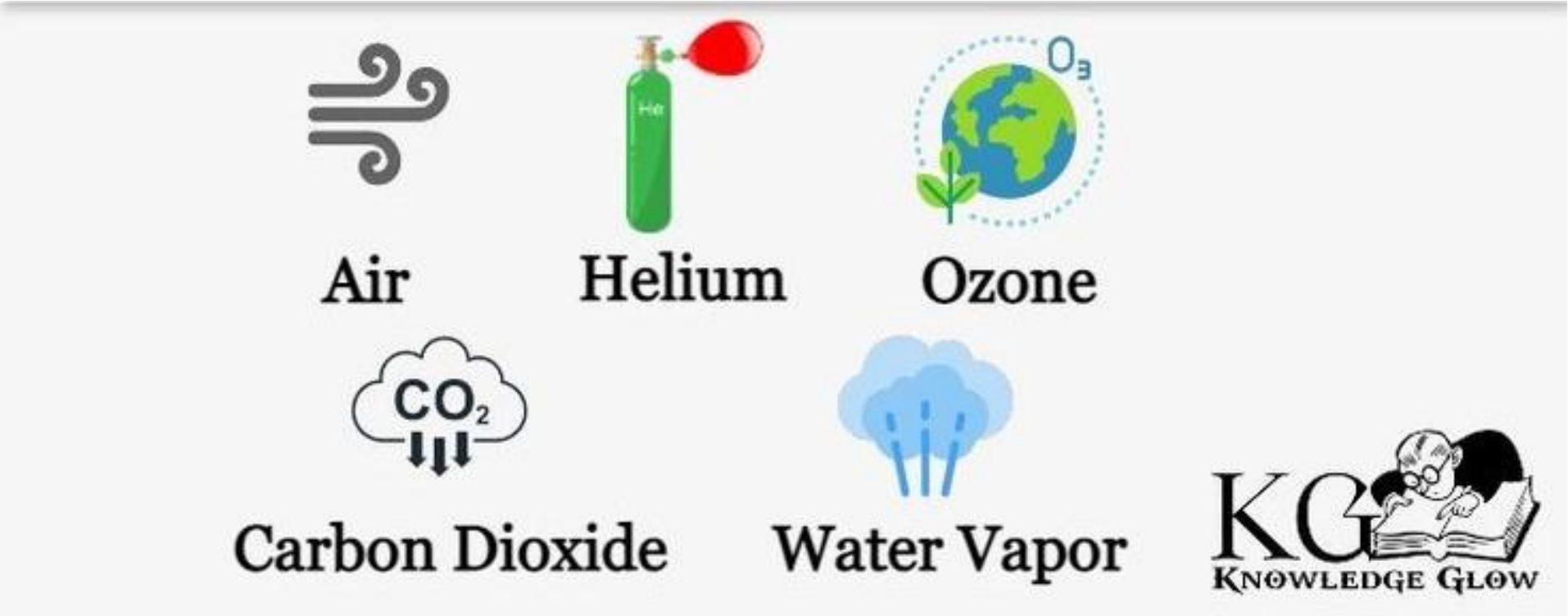
- Not Rigid
- No Fixed Shape
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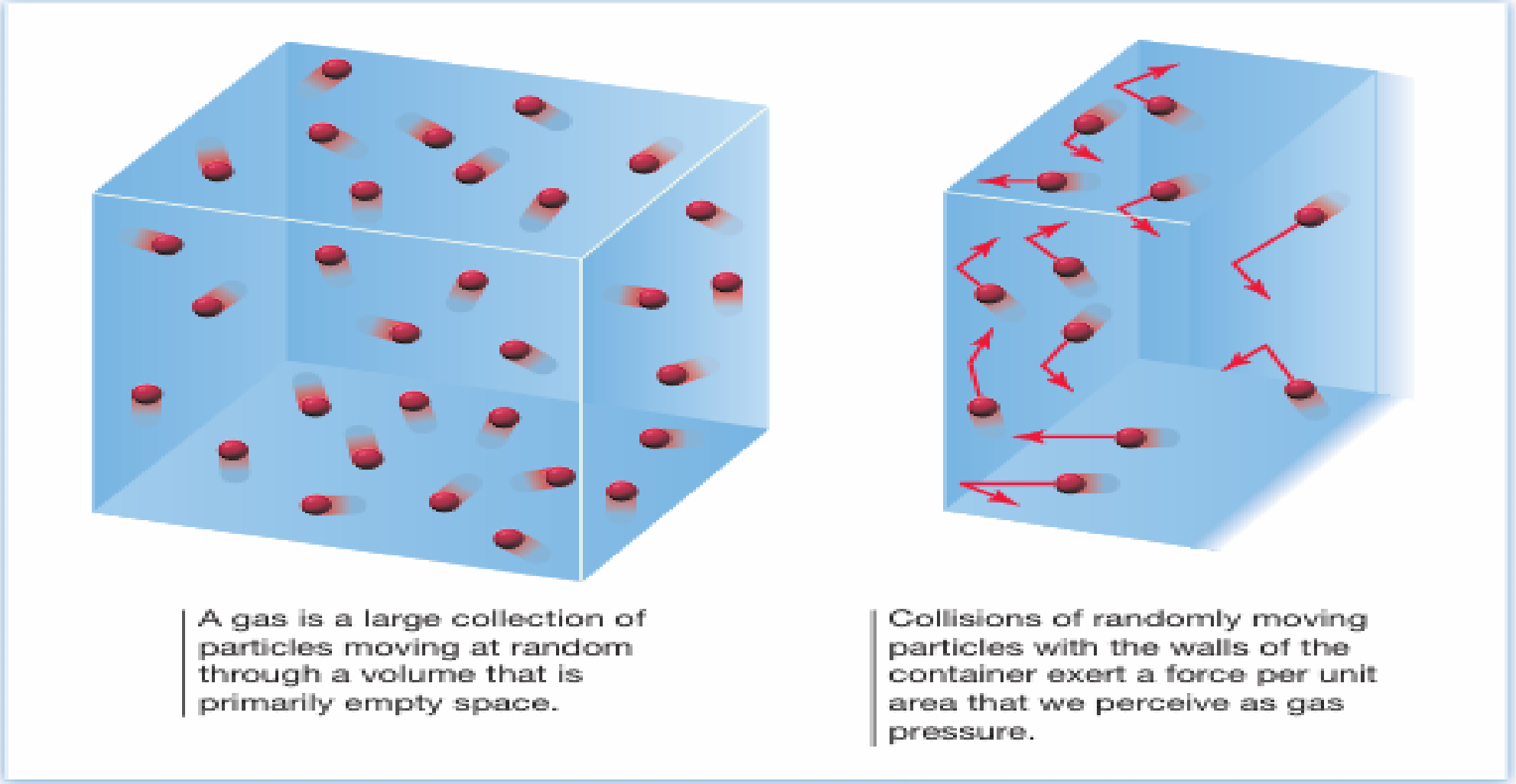
- **Solids:** are substances whose atoms, molecules, or ions are held rigidly together, giving a **defined volume and shape**.
- **Liquids:** atoms or molecules are held together less strongly, giving the liquid a **definite volume** but a **changeable and indefinite shape**
- **Gases:** whose atoms or molecules have a weak attraction. Expand freely to fill the whole of a container, having an **undefined shape or volume**

Part 1:Gases
Their Properties and Behavior

1.1. Gases and gas pressure

Gases can exist as **single atoms** as **noble gases like helium (He)**, or **single elements**, like **hydrogen gas (H_2),** a compound, like **carbon dioxide (CO_2),** or a **mixture of several gases**, like **air or natural gas**.





A gas is a large collection of particles moving at random through a volume that is primarily empty space.

Collisions of randomly moving particles with the walls of the container exert a force per unit area that we perceive as gas pressure.

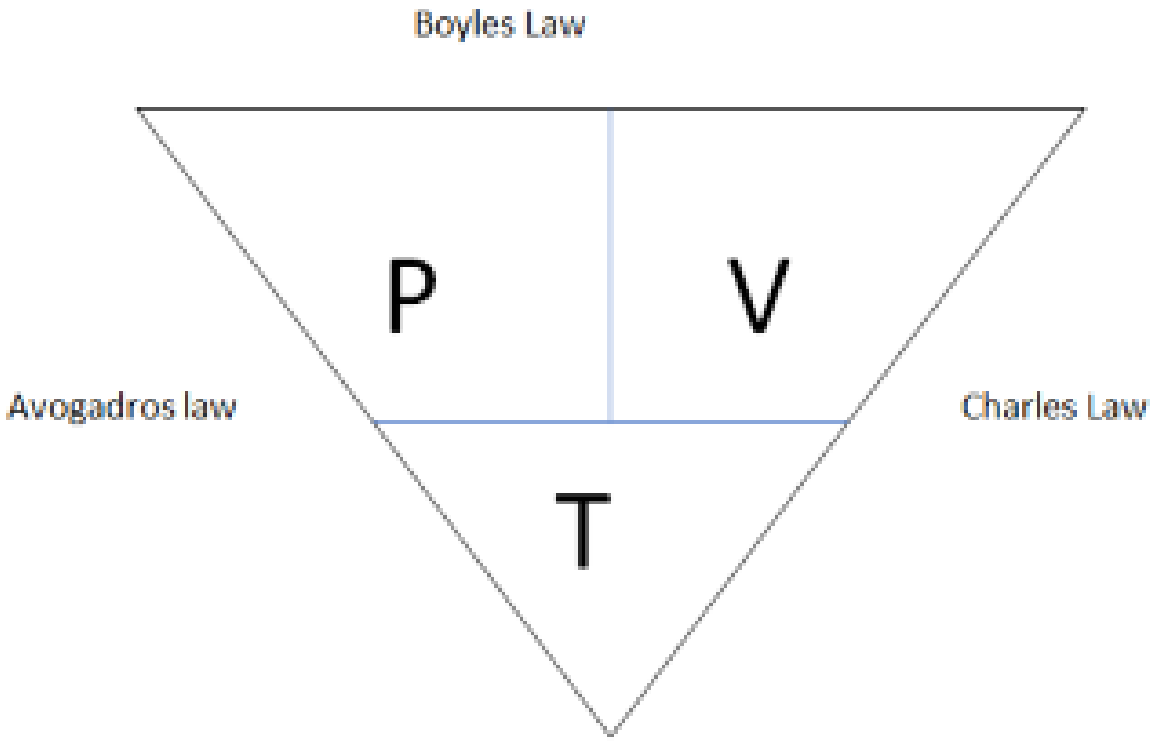
Note: In gases **less than 0.1%** of the volume of a typical gas is taken up by the molecules themselves under ideal circumstances **(high temperature and low pressure)**

1.2. The gas laws

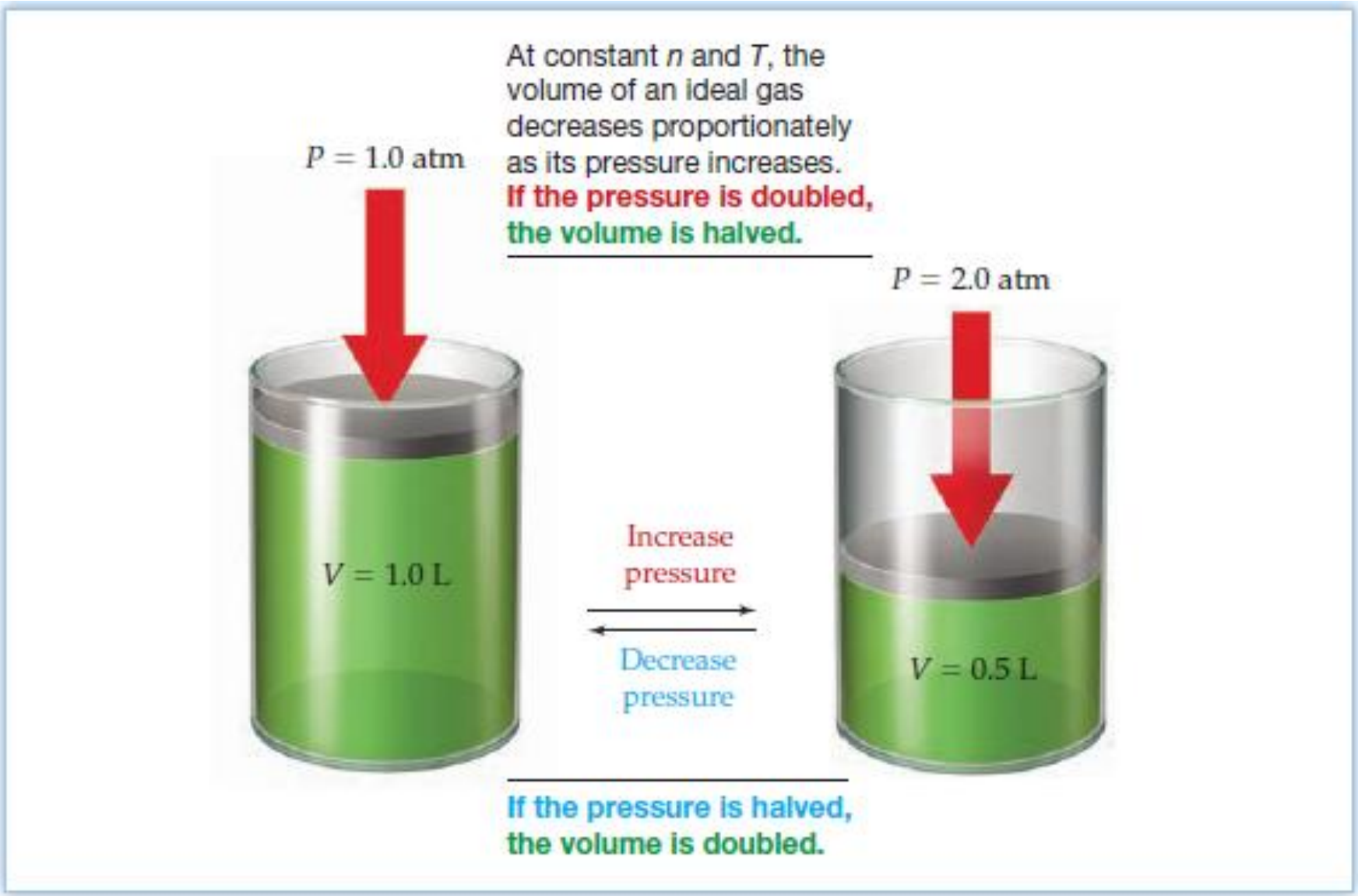
- **Boyle's Law:** The Relationship between Gas Volume and Pressure

- **Charles's Law:** The Relationship between Gas Volume and Temperature

- **Avogadro's Law:** The Relationship between Volume and Amount

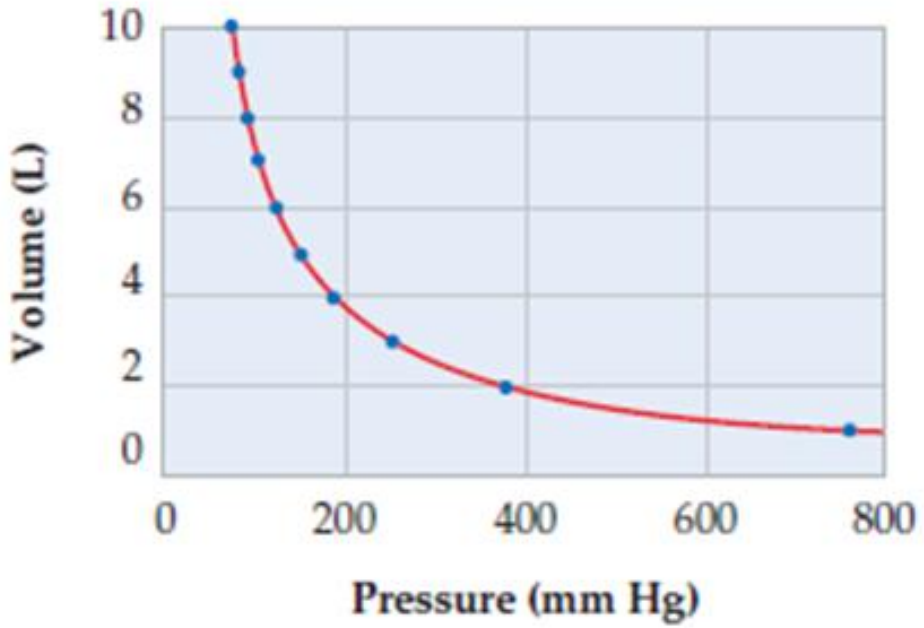


Boyle's Law

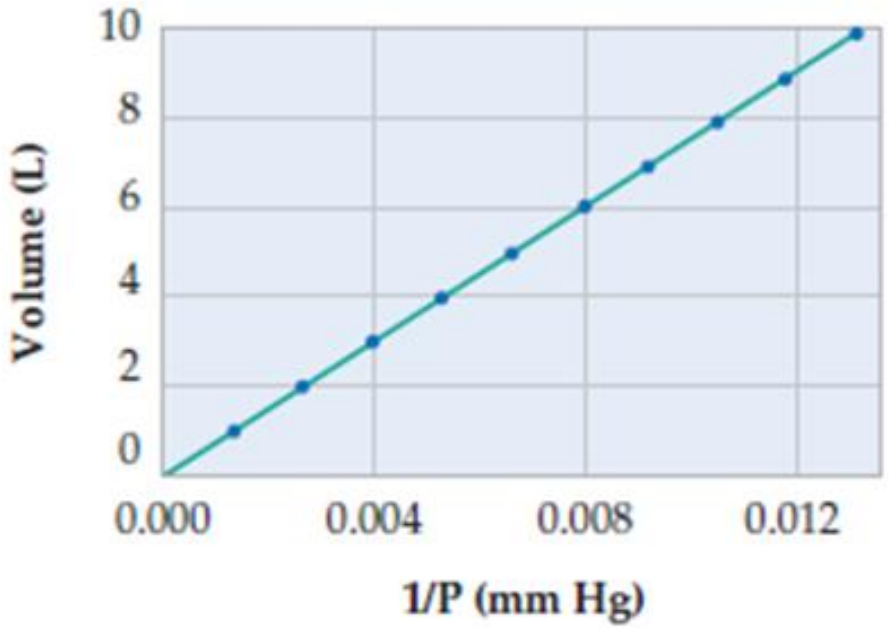

At constant n and T, the volume of an ideal gas decreases proportionately as its pressure increases.
If the pressure is doubled, the volume is halved.
P = 1.0 atm
V = 1.0 L
Increase pressure
Decrease pressure
P = 2.0 atm
V = 0.5 L
If the pressure is halved, the volume is doubled.

Boyle's Law

$V \propto 1/P$ or $PV = \text{Constant}$ at fixed n and T

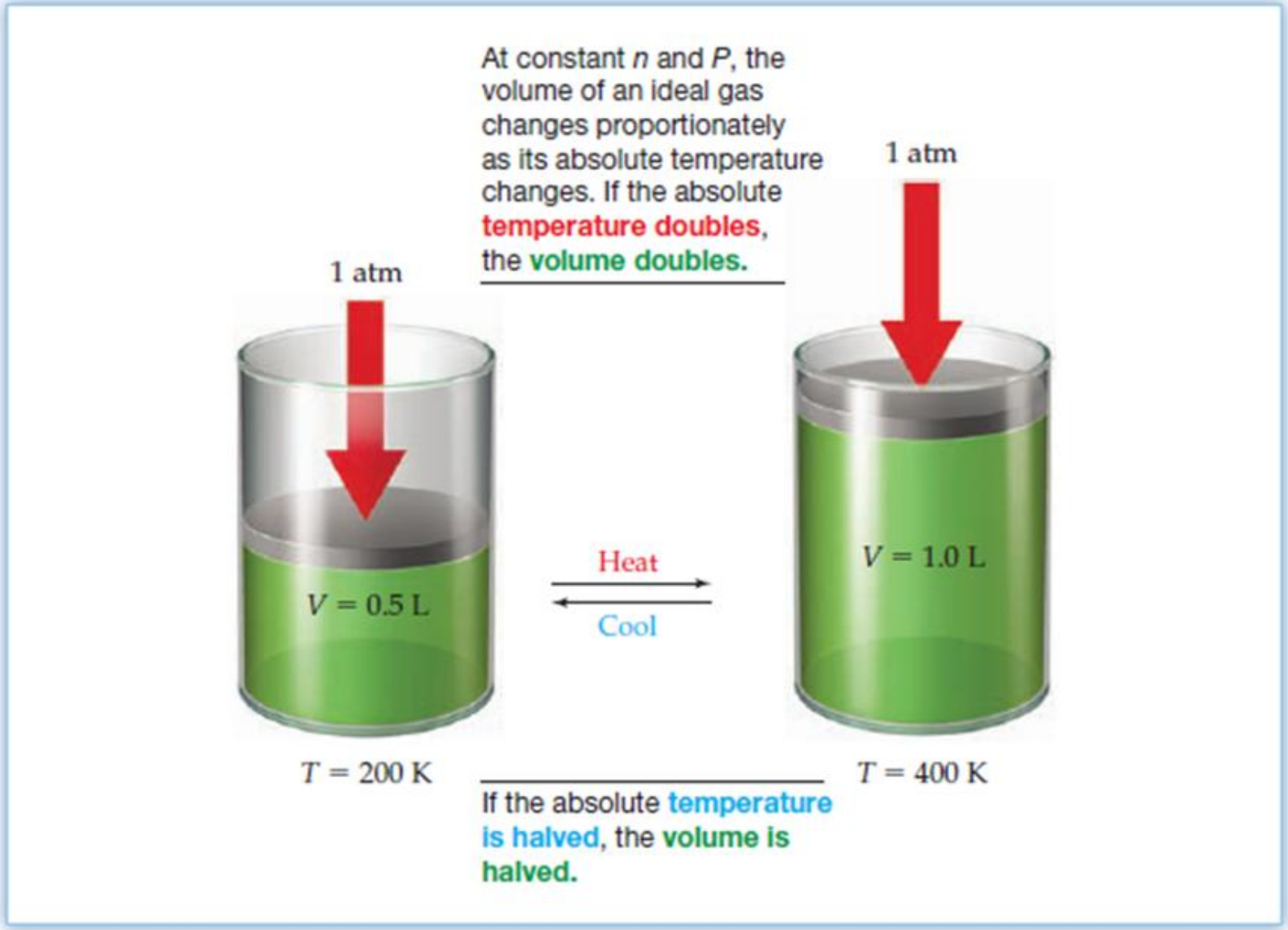


A plot of *V* versus *P* for a gas sample is a **hyperbola**.



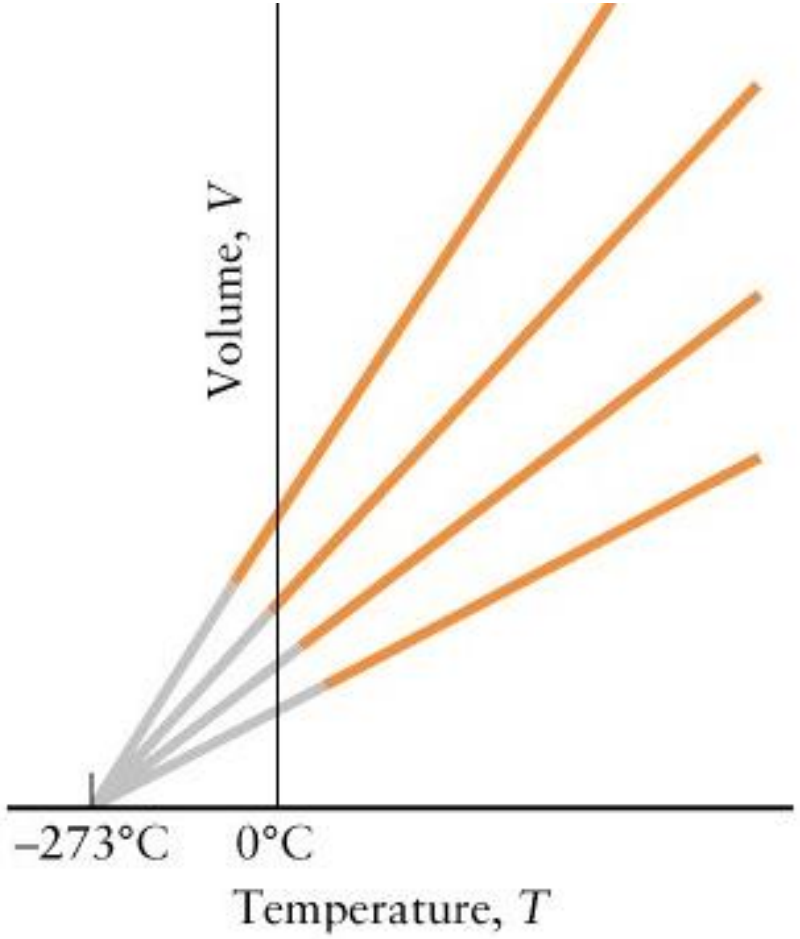
A plot of *V* versus 1/*P* is a **straight line**. Such a graph is characteristic of equations having the form $y = mx + b$.

Charles's Law

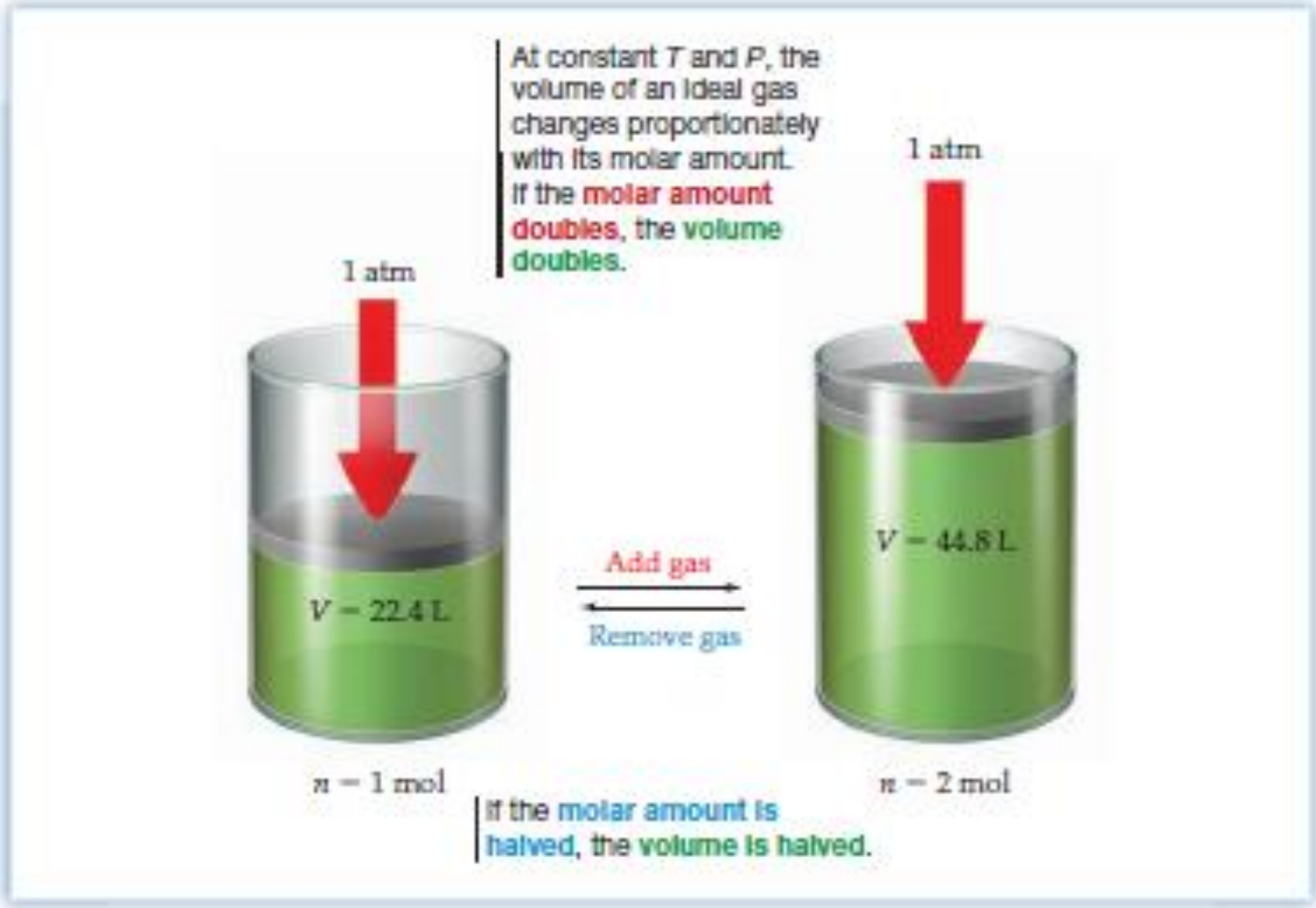


Charles's Law

$V \propto T$ or $V/T = \text{Constant}$ at fixed n and P

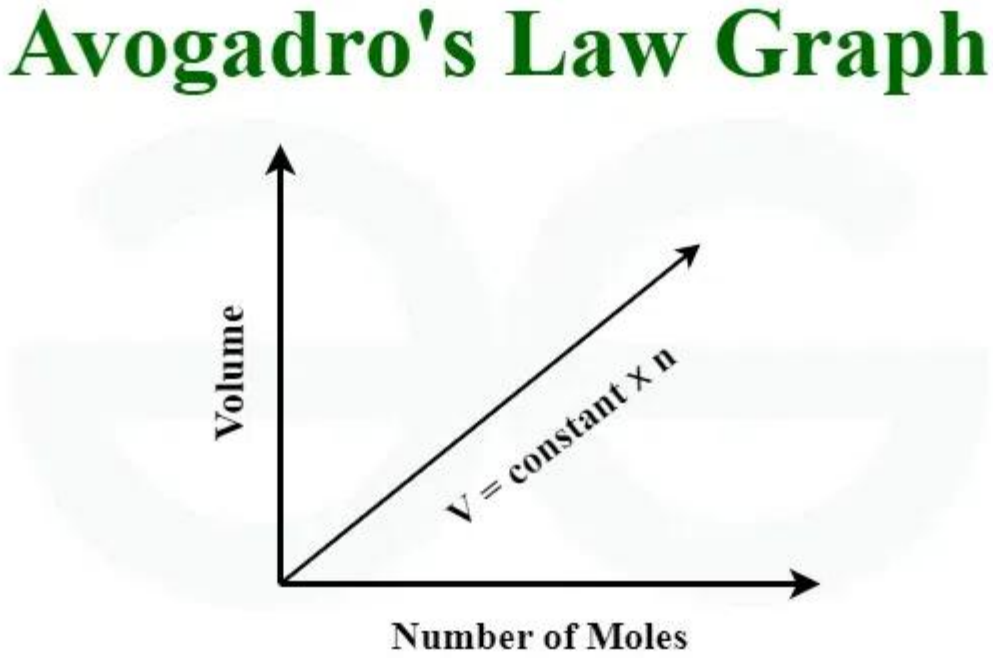


Avogadro's law

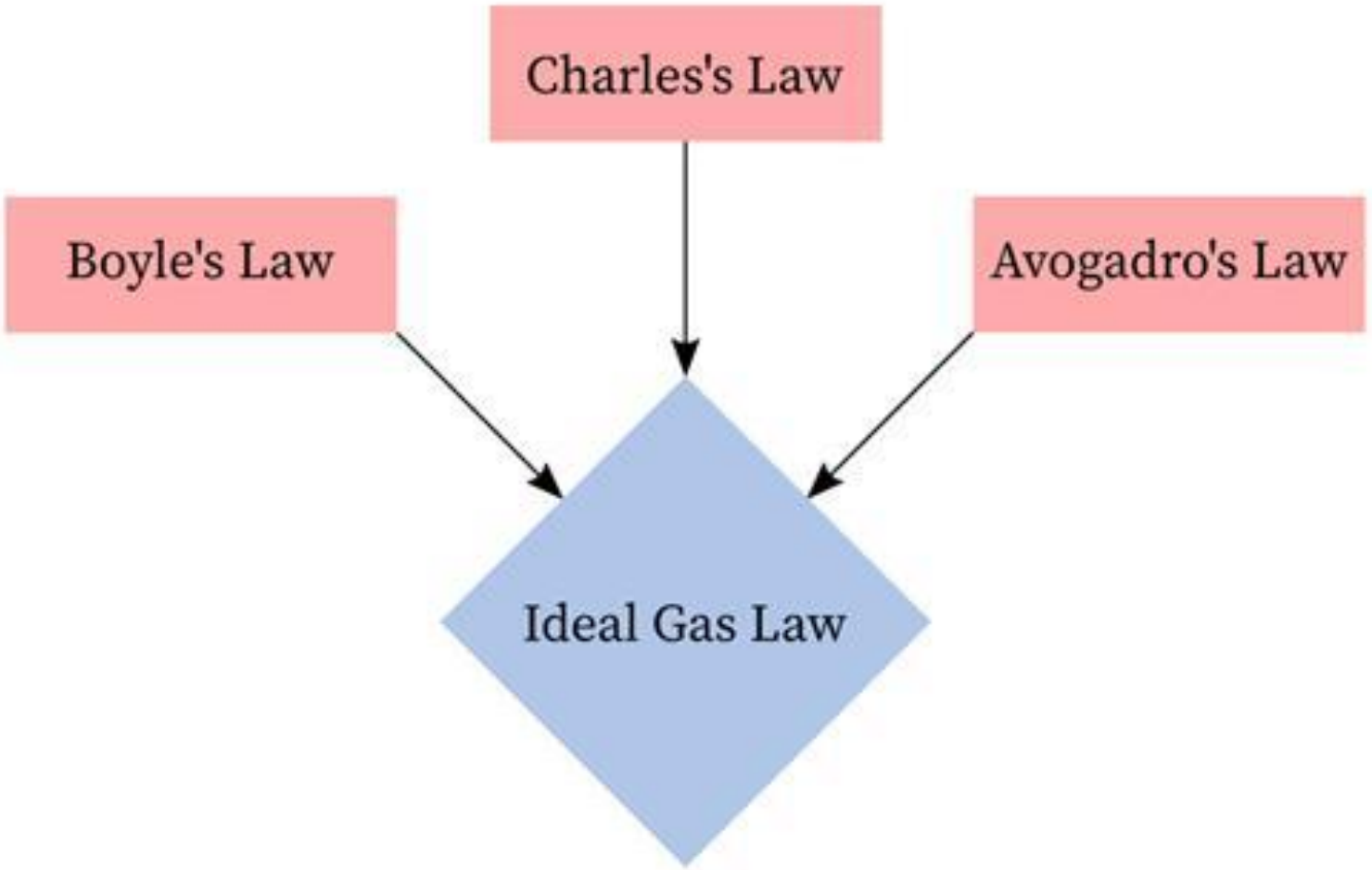


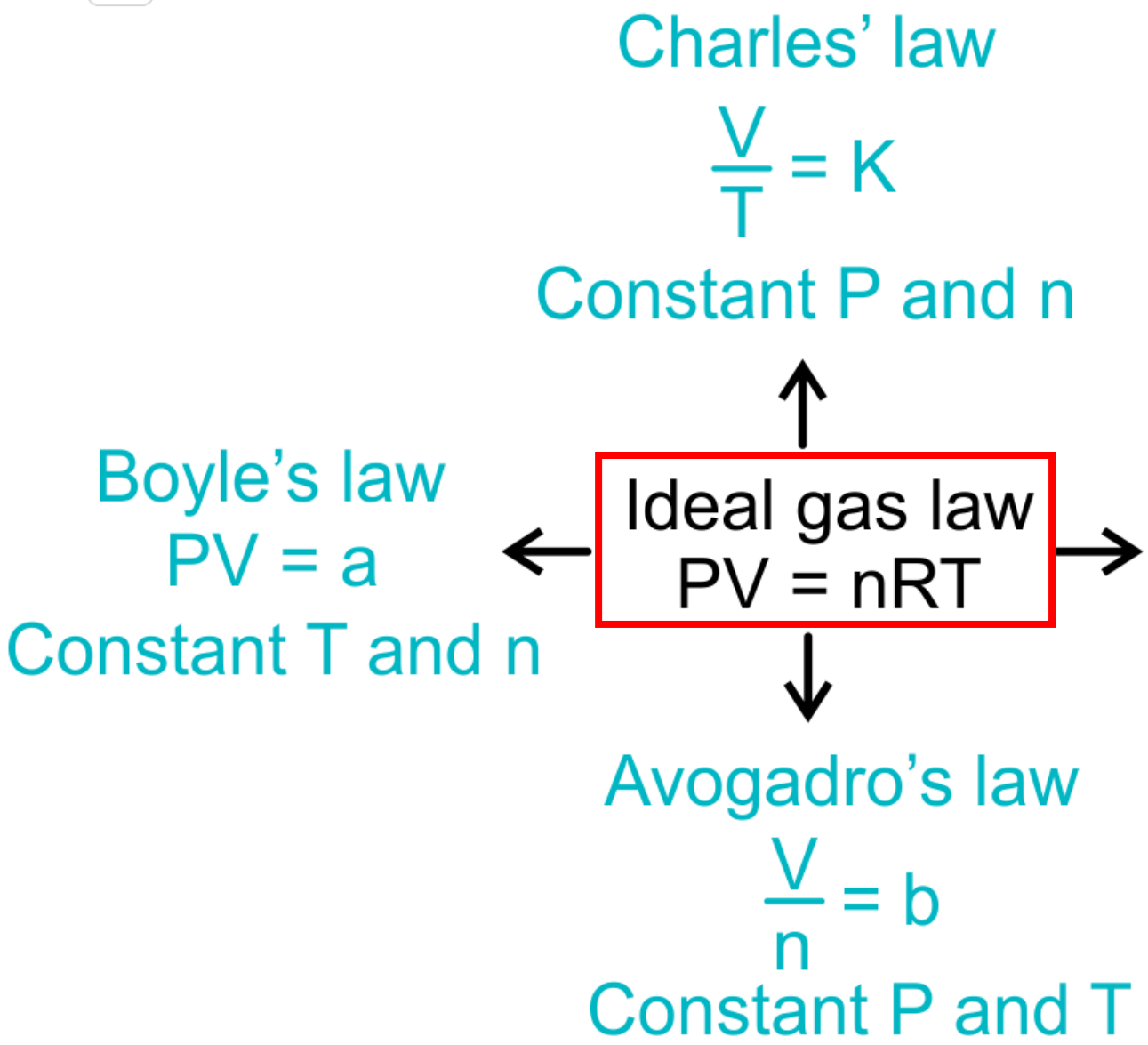
Avogadro's law

$$V \propto n \quad or \quad \frac{V}{n} = Constant\ at\ fixed\ T\ and\ P$$



1.3. The ideal gas law





Where:

- P is the pressure of the gas,
- V is the volume of the gas,
- n is the amount of substance of gas (also known as number of moles),
- R is the ideal, or universal, gas constant
- T is the absolute temperature of the gas

The ideal gas equation can be arranged to reach another form

$$PV = nRT \div m \quad where\ m\ is\ the\ gas\ weight$$

$$P\frac{V}{m} = \frac{n}{m}\ RT$$

$$\frac{V}{M} = \frac{1}{\rho} \quad and\ \frac{n}{m} = \frac{1}{M}\ where\ M\ is\ the\ molar\ mass$$

$$so\ P\frac{1}{\rho} = \frac{1}{M}\ RT \quad \Rightarrow PM = \rho RT$$

$$\boxed{\rho = \frac{PM}{RT}}$$

Avogadro's hypothesis







Volume	22.4 L	22.4 L	22.4 L
Pressure	1 atm	1 atm	1 atm
Temperature	0 °C	0 °C	0 °C
Mass of gas	4.00 g	28.0 g	16.0 g
Number of gas molecules	6.02×10^{23}	6.02×10^{23}	6.02×10^{23}

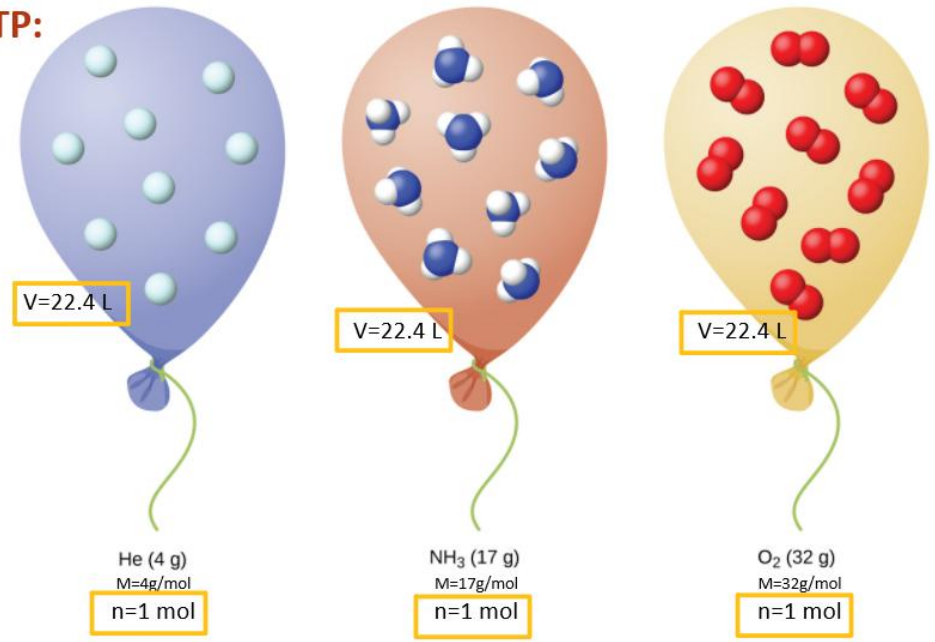
STP

STANDARD TEMPERATURE and PRESSURE

P = 1 atm
= 760 mmHg
=101.3 kPa

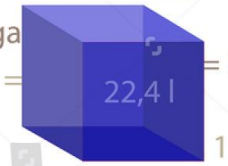
T = 273 K
= 0°C

@ STP:



Vm

Molar volume
is a physical quantity



32g of O_2
(1 mol)



2g of H_2
(1 mol)

$$V_m = \frac{V}{n}$$

Vm = 22,4 l/mol

1 mol of gas = 22,4 L

Universal gas constant [R] calculation

$$R = \frac{P.V}{n.T} = \frac{(1\ atm)(22.414\ L)}{(1\ mol)(273.15\ K)} = 0.082058\ \frac{L.atm}{K.mol}$$

R = 8.3145 J/(K.mol) in SI units

Solved examples

In a typical automobile engine, gasoline and air mixture is compressed from 1.0 to 9.5 atm in a cylinder before ignition. If the uncompressed volume of the cylinder is 410 mL, what is the volume in milliliters when the mixture is fully compressed?

Solution

$$\mathrm{PV}_{final} = PV_{Initial} = nRT$$

$$V_{final} = \frac{PV_{Initial}}{P_{final}} = \frac{1.0\ atm \times 410\ ml}{9.5\ atm} = 43\ ml$$

Solved examples

Oxygen gas is commonly sold in 49.0 L steel containers at a pressure of 150 atm and temperature of 20 °C. What volume in liters would the gas occupy

- At a pressure of 1.02 atm if its temperature remained unchanged?
- If its temperature was raised from 20.0 °C to 35.0 °C at constant pressure equals 150 atm?

Solution

- $\frac{V_1}{V_2} = \frac{P_2}{P_1} = \frac{49\ L}{V_2} = \frac{1.02\ atm}{150\ atm} \quad \Rightarrow V_2 = \frac{49 \times 150}{1.02} = 7205.88\ L$

- $\frac{T_1}{T_2} = \frac{V_1}{V_2} = \frac{(20+273.15)\ K}{(35+273.15)K} = \frac{49.0\ L}{V_2} \Rightarrow V_2 = 51.51\ L$

Solved examples

A typical high-pressure tire on a bicycle might have a volume of 365 mL and a pressure of 7.80 atm at 25 °C. Suppose the rider filled the tire with helium to minimize weight What is the mass of the helium in the tire (He atomic weight =4)?

Solution

$$\mathrm{n} = \frac{\mathrm{P} \times \mathrm{V}}{\mathrm{R} \times \mathrm{T}} = \frac{7.8\ atm\ \times 0.365\ L}{0.082\ \frac{L.atm}{K.mol} \times 298\ K} = 0.116\ mol$$

$$\text{Helium mass} = 0.116\ \text{mol} \times 4\ \frac{\text{g}}{\text{mol}} = 0.464\ \text{g}$$

Solved examples

To identify the contents of an unlabeled cylinder of gas, a sample was collected and found to have a density of 5.380 g/L at 15 °C and 736 mm Hg pressure. What is the molar mass of the gas?

Solution

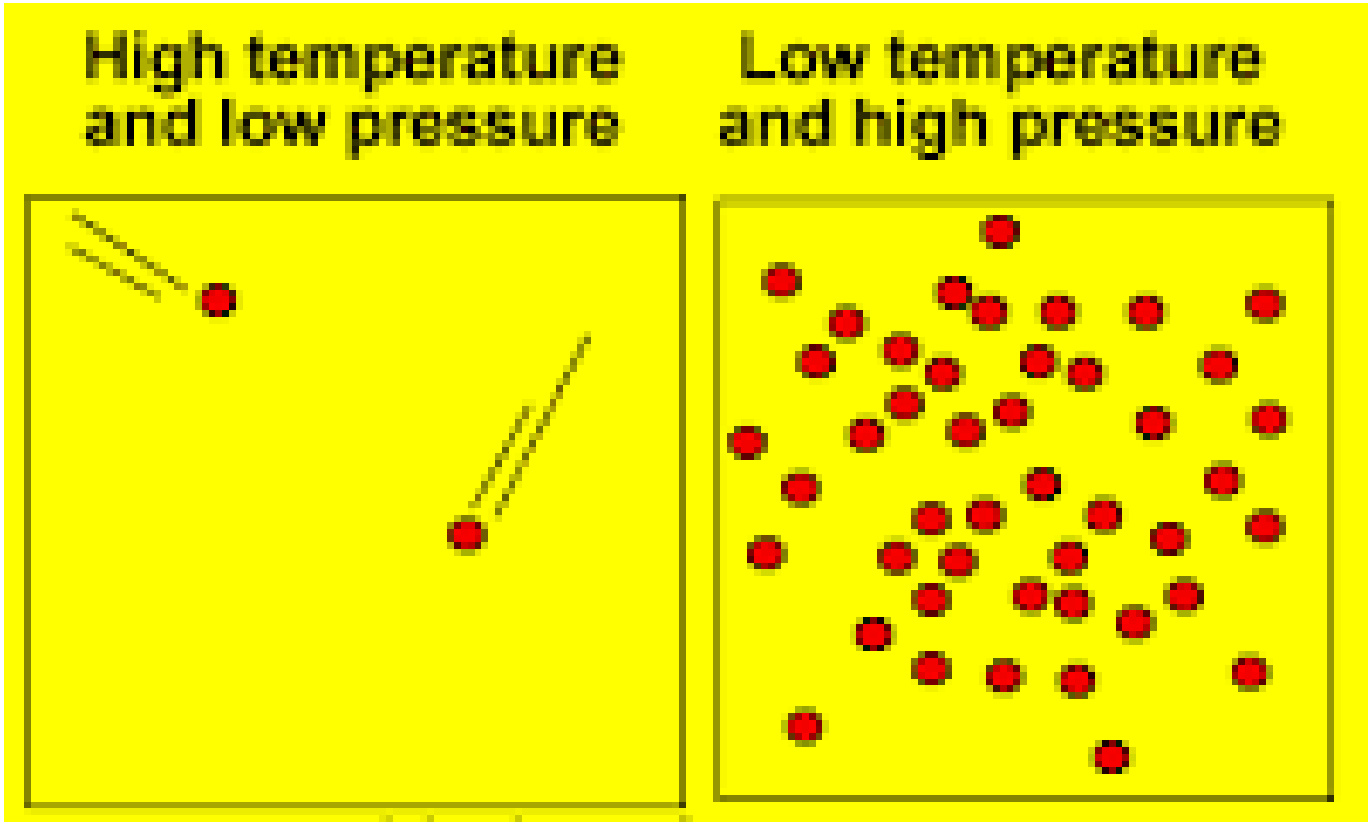
1 atm =760 mmHg so 736 mmHg = 0.968 atm

ρ=PM/RT $5.38 = \frac{[0.968 \times M]}{[0.082 \times (15+273)]} \Longrightarrow M =$

$\frac{5.38 \times 0.082 \times 288}{0.968} \approx 131.2\ g/mol$

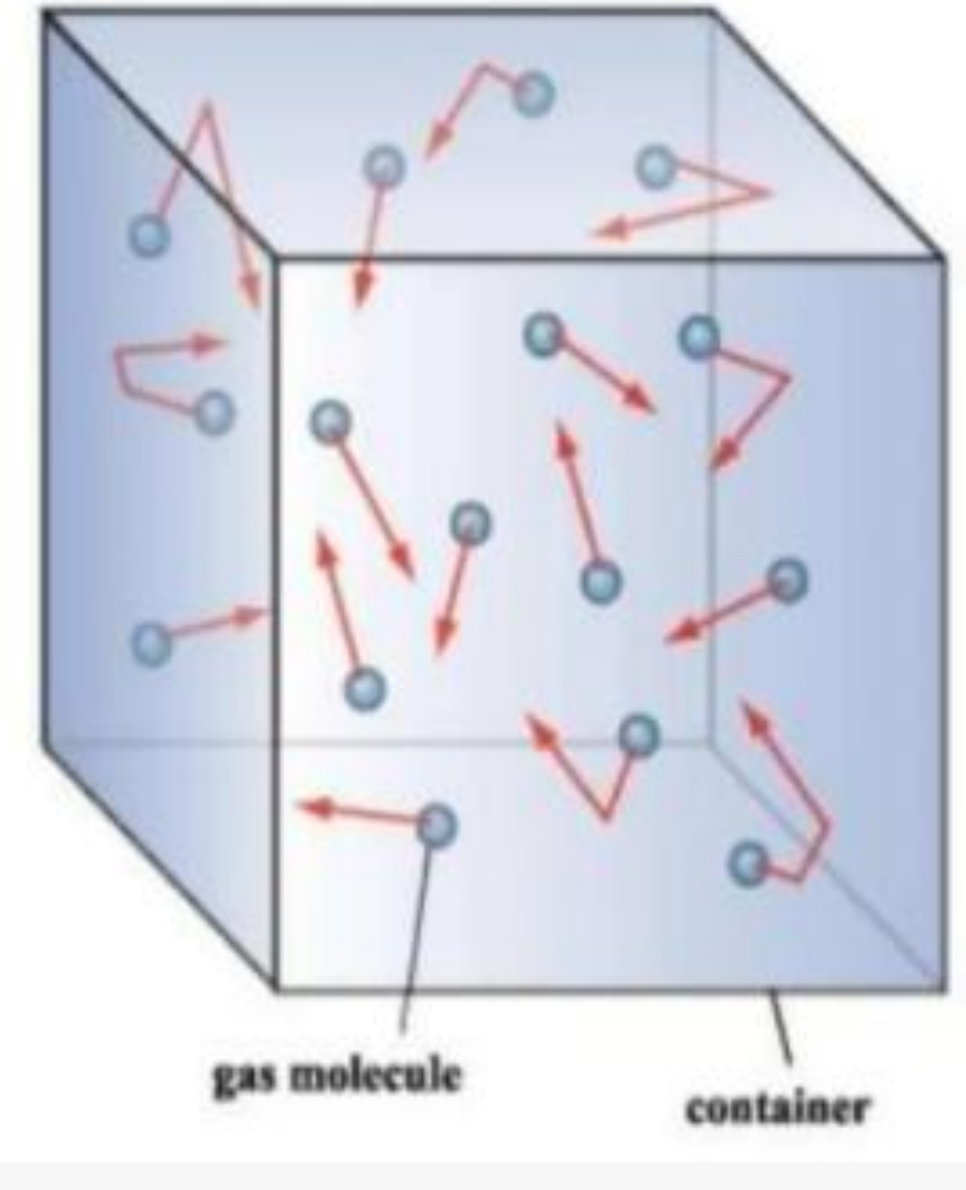
1.4.The kinetic molecular theory of gases

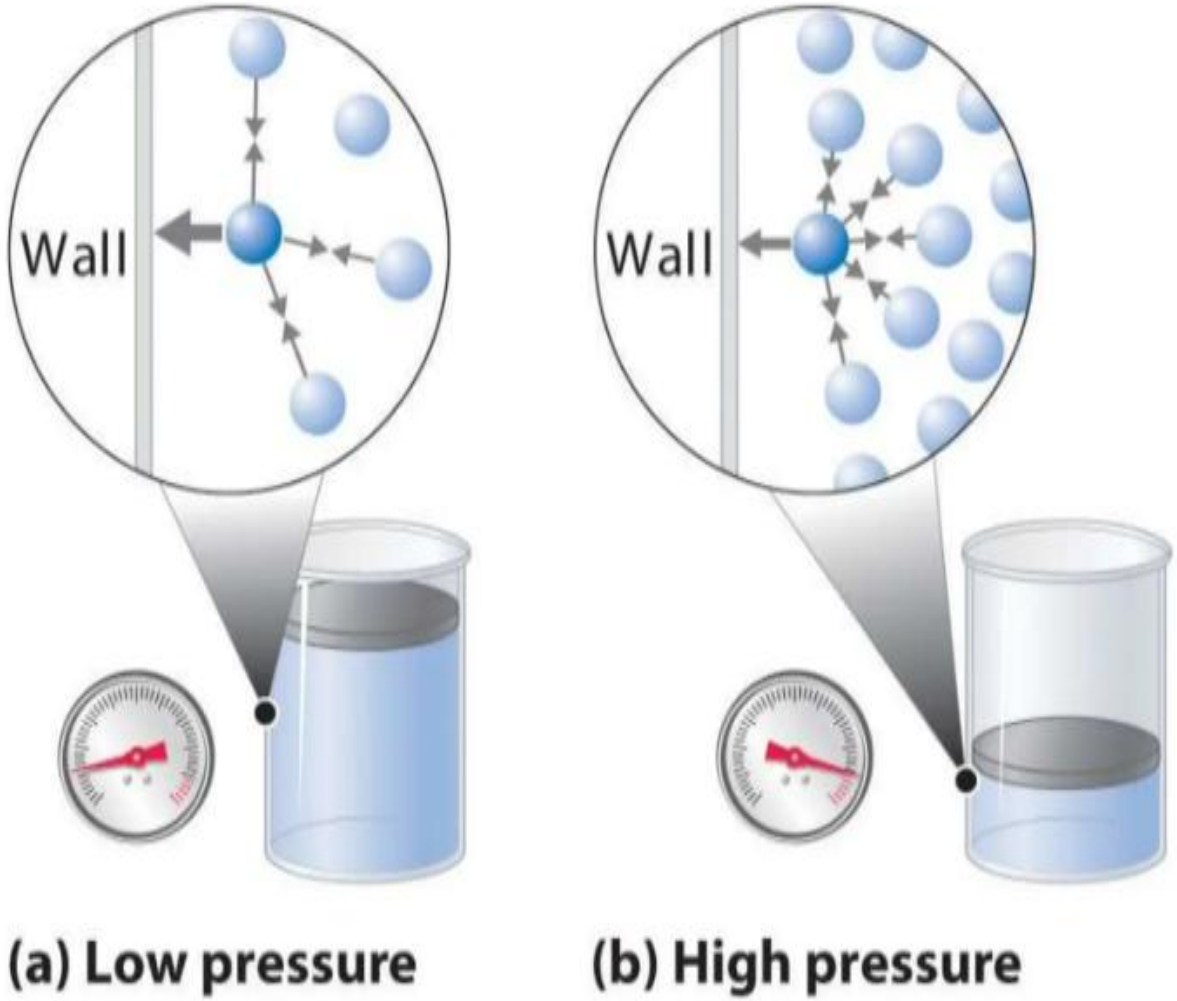
A model that helps us explain what happens to gas particles when conditions such as pressure or temperature change.

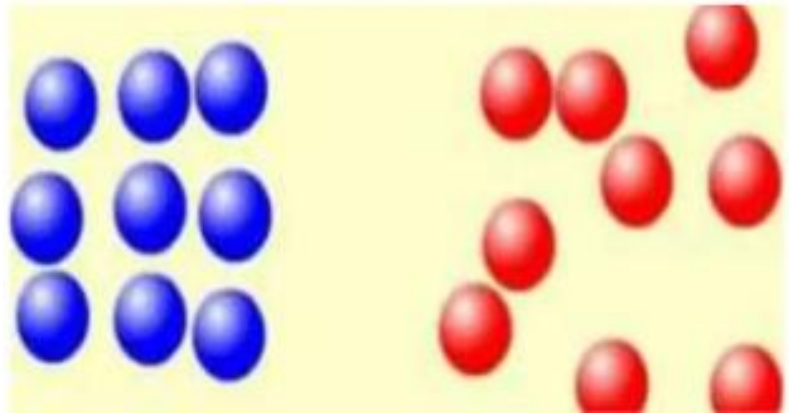


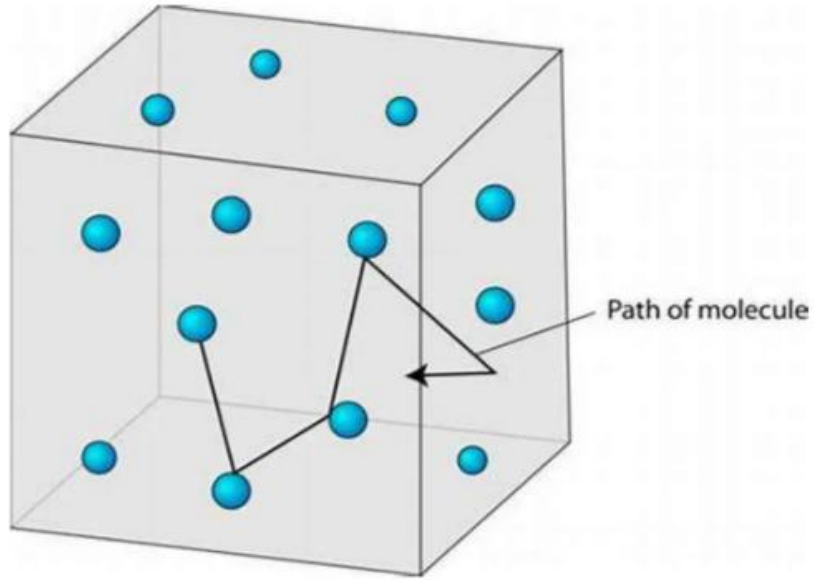
- Gases consist of large numbers of molecules that are in continuous random motion.

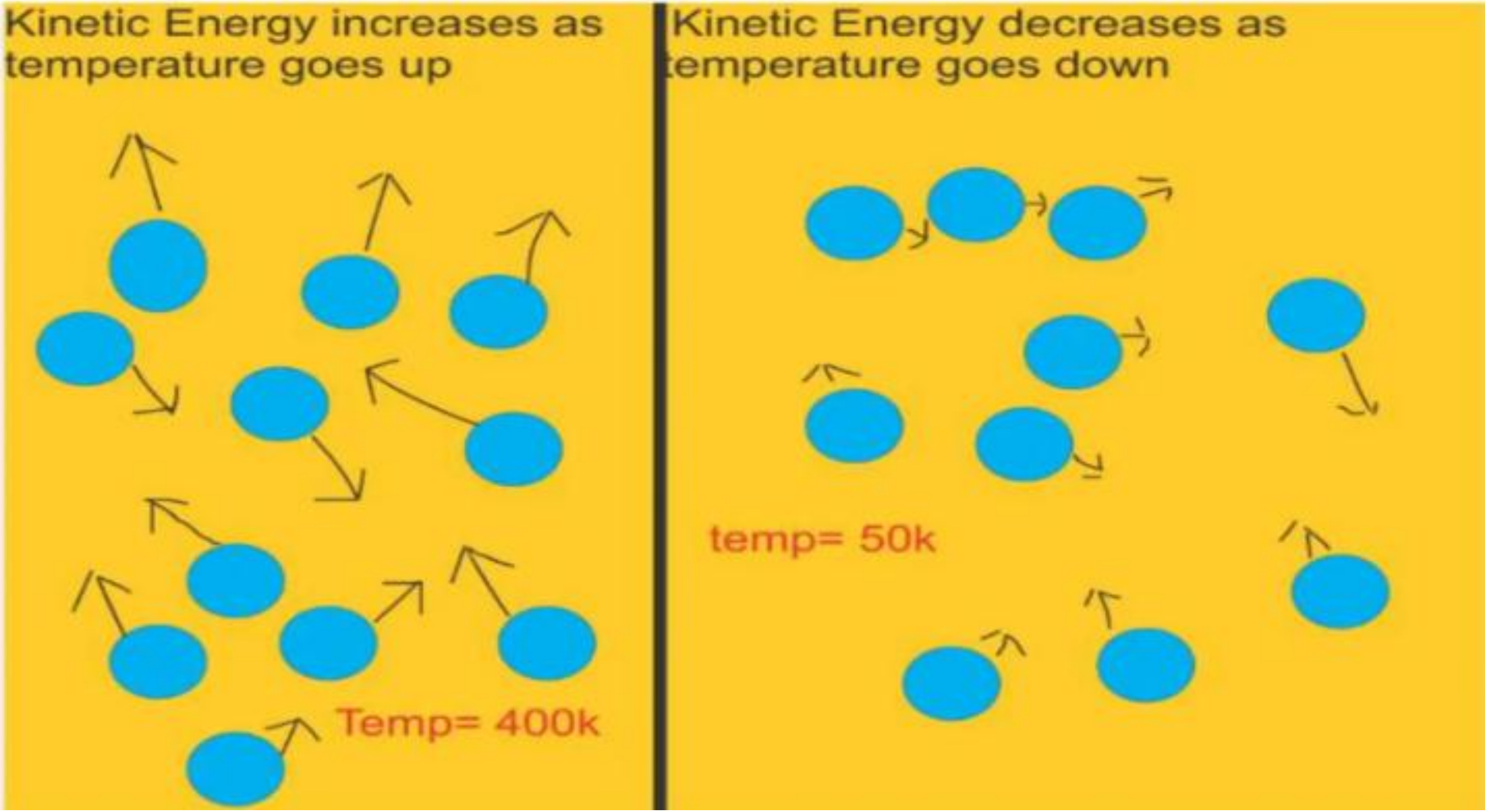
- The combined volume of all the gas molecules is negligible relative to the total volume of the gas.
- Attractive and repulsive forces between gas molecules are negligible.
- Energy can be transferred between molecules during collisions but, as long as temperature remains constant, the average kinetic energy of the molecules does not change with time.




Wall
Wall
(a) Low pressure
(b) High pressure




Path of molecule


Kinetic Energy increases as temperature goes up
Temp= 400k
Kinetic Energy decreases as temperature goes down
temp= 50k

1.5. The behavior of real gases





- Attractive and repulsive forces between gas molecules are can not negligible.

- The combined volume of all the gas molecules can not negligible relative to the total volume of the gas.

Correction for intermolecular attractions

$$[P + (an^2/V^2)]\,[V - nb] = nRT$$

Correction for molecular volume

Or

$$P = \frac{nRT}{V - nb} - \frac{an^2}{V^2}$$

Where: a L^2atm/mol^2

b L/mol

Solved example

Assume that you have 0.5 mol of N_2 in a volume of 0.6 L at 300 K. Calculate the pressure in atmospheres using both the ideal gas law and the Van der Waals equation. For N_2

(a =1.35 (L^2.atm)/mol^2, and b= 0.0387 L/mol) .

Solution

- **Using ideal gas law**

$$P = \frac{nRT}{V} = \frac{0.5\ mol \times 0.082\left(\frac{atm.L}{mol.K}\right) \times 300\ K}{0.6\ L} = 20.5\ atm$$

Solution

- **Using real gas law**

$$P = \frac{nRT}{V - nb} - \frac{an^2}{V^2}$$

$$P = \frac{0.5mol \times 0.082\left(atm.L/_{mol.K}\right) \times 300\,K}{0.6\,L - 0.5\,mol \times 0.0387\left(L/_{mol}\right)} - \frac{1.35\left(L^2.atm/_{mol^2}\right) \times 0.5^2\,mol^2}{0.6^2\,mol^2}$$

$$= 20.25\,atm$$